

Root relationships-Quadratic

Key Points

Then the two roots, α, β have some nice properties

$$\alpha + \beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

Manipulate these to find related root equations, the choice of a is arbitrary.

If the roots are symmetrically alerted you can instead replace a variable using substitution

Basic Skills

Q1. Solve the following quadratic equations

a) $3x^2 + 6x - 9 = 0$

b) $2x^2 - 10x + 6 = 0$

Q2. For the quadratic equation $2x^2 + 8x - 10 = 0$

a) Show that the roots are $\alpha = 1$ and $\beta = -5$

b) Verify that $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$

c) Show that the quadratic equation can be written as $(x - \alpha)(x - \beta) = 0$

d) By replacing α, β with $\alpha + 1, \beta + 1$ find a quadratic equation with roots of 2 and -4

Competency Skills

Q3. The quadratic equation $3x^2 + 14x - 3 = 0$ has roots α, β

a) Use $\alpha + \beta = -\frac{b}{a}$, $\alpha\beta = \frac{c}{a}$ to form a simultaneous equation for the roots

b) Hence find α, β and state what you notice

c) Hence find state a quadratic equation with roots $2\alpha, 2\beta$

Q4. The quadratic equation $2z^2 + 5z + 10 = 0$ has roots α, β

a) Find a quadratic equation with roots $\alpha + 1, \beta + 1$

b) Show that replacing z with $z + 1$ in the original equation produces the new quadratic equation in a)

c) Find the related quadratic equation with roots $2\alpha + 1, 2\beta + 1$

Advanced Skills

Q5. The quadratic equation $4z^2 + 3z - 1 = 0$ has roots α, β

a) Find the quadratic equation with roots $2\alpha^{-1}, 2\beta^{-1}$

b) Find the quadratic equation with roots $\frac{\alpha-1}{\beta}, \frac{\beta-1}{\alpha}$

Q6. The quadratic equation $z^2 - kz + 6 = 0$ has roots $\alpha, \alpha + 1$ find two possible values of k

Q7. Without solving the quadratic equation $z^2 + 5z + 10 = 0$, which has roots α, β , find the value of $\alpha^3 + \beta^3$

Q8. The quadratic equation $z^2 - z - 5 = 0$ has roots α, β , find a quadratic equation with roots $\alpha^2 + 1, \beta^2 + 1$

Extension

Q9. Can you find a rule for the cubic equation with roots α, β, γ

Hint: Look at the cubic $(z - \alpha)(z - \beta)(z - \gamma) = 0$ and expand to $az^3 + bz^2 + cz + d = 0$

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